Extended Syllabus

Course Title	Research Method of Data Science	**Semester**	Fall 2026
Credit	3	**Course Number**	TIC3002
Class Time	Tue/Thu 13:30-14:45	**Enrollment Eligibility**	Primarily 3rd-year GSIS students; open to qualified students

Instructor	**Name: Iegor Vyshnevskyi**	Homepage: https://iegor-vy.github.io/
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Ⅰ. Course Overview

1. Description

This course teaches students how to turn substantive questions in economics, business, public policy, and international affairs into credible, reproducible data-science research. Students work through the full research cycle: framing a question, translating concepts into measurable variables, locating and documenting data, cleaning and joining datasets, exploring patterns, selecting an appropriate research design, estimating and checking models, and communicating results.

The course uses different research methods and types of evidence to teach students how to conduct a complete study. It therefore includes literature searching and synthesis, qualitative research, quantitative research design and analysis, and research communication. Python is used as the primary analytical language in Google Colab, with GitHub supporting versioned and reproducible work. Generative AI and coding assistants are taught explicitly as research tools, but all AI-assisted work must be disclosed, tested, sourced, and defended by the student. By the end of the semester, each student will complete and present an auditable individual research project addressing a real global or social-science problem.

2. Prerequisites

Students should have completed at least one university-level statistics course and should be comfortable with basic algebra, functions, graphs, and introductory economic reasoning. Prior programming experience is helpful but not required. This is not a Python-from-zero course; short preparatory materials and starter notebooks will support students who need to refresh essential coding skills.

3. Course Format (%)

Lecture	Discussion	Experiment /Practicum	Field study	Presentations	Other
30 %	20 %	40 %	0 %	10 %	0 %

4. Evaluation (%)

Mid-term Exam	Final exam	Quizzes	Presentations	Group Activities	Individual Assignments	Participation	Other
25 %	35 %	0 %	0 %	20 %	10 %	10 %	0 %

Ⅱ. Course Objectives

By the end of the course, students will be able to:

1. Formulate a feasible research question and connect it to theory, a clearly identified literature gap, appropriate evidence, and a defensible research design.

2. Search for, evaluate, organize, and synthesize academic literature; distinguish summary from critical synthesis; and develop a coherent conceptual framework.

3. Explain the purposes, strengths, limitations, and ethical requirements of major qualitative and quantitative approaches and select methods that fit the question.

4. Design and conduct a small qualitative inquiry using interviews, observation, documents, or case evidence, and analyze qualitative material through transparent coding.

5. Find, document, clean, join, and validate real-world data and use Python for reproducible visualization, inference, regression, and selected advanced applications.

6. Distinguish description, association, prediction, and causal explanation; interpret evidence with appropriate uncertainty; and recognize threats to validity.

7. Use AI tools responsibly by disclosing assistance, verifying code and claims, checking sources, and retaining ownership of all submitted reasoning.

8. Write and defend a complete individual study containing an introduction, literature review, data and methods, results, discussion, conclusion, and limitations, supported by a well-organized repository and reproducible analytical materials.

Ⅲ. Course Format

(* In detail)

Each week normally combines two linked sessions:

- Tuesday: concepts, research design, worked examples, and guided discussion.

- Thursday: research studio with literature, qualitative evidence, or real data; short demonstrations; individual practice; structured peer feedback; and project development.

Google Colab is the primary computing environment. GitHub is used to distribute materials and to organize reproducible individual work. Cyber Campus remains the official channel for announcements, deadlines, and required submissions.

The course follows a scaffolded individual-project model. Students develop one study through linked milestones: question, literature map, proposal, qualitative exercise, data and methods, analysis, full draft, presentation, and final submission. Early labs isolate individual skills; later work combines them into a coherent research argument.

IV. Course Requirements and Grading Criteria

Assessment is based on demonstrated research competence:

- Research workflow labs participation: 10% (individual; literature searching, data handling, visualization, statistical reasoning, and reproducibility checks).

- Literature review and research proposal: 15% (group/individual; research question, search strategy, critical synthesis, literature gap, conceptual framework, and design).

- Midterm practical assessment: 25% (individual).

- Qualitative research exercise: 10% (group/individual; research design, evidence collection or document-based inquiry, coding, interpretation, reflexivity, ethics, and limitations).

- Final examination assignment - individual research project: 35% (complete paper, reproducible supporting materials and presentation). The paper must include an introduction, literature review, data and methods, results, discussion, conclusion, and limitations. Individual presentation of the complete study, followed by questions about the literature, design, evidence, and findings. Prepared participation and structured peer review.

Unless stated otherwise, assignments are due before the beginning of class on the due date. Late work and revisions follow the instructions posted for each assignment.

V. Course Policies

Academic integrity and authorship

Students may use generative AI or coding assistants only within the conditions stated for each task. Permitted use must be disclosed. Students must retain a brief AI-use record, verify code and factual claims, cite original sources, and be able to explain and reproduce every submitted result. Fabricated data, references, outputs, or concealed AI-generated work constitute academic misconduct. Literature citations suggested by AI must be located, read, and verified against the original publication. AI tools are not permitted during the midterm practical assessment unless explicitly authorized.

Reproducibility

Submitted work must run from the stated starting point, use documented data sources, preserve raw data where licensing permits, and clearly distinguish inputs, transformations, analysis, and outputs. Non-reproducible results may receive substantially reduced credit.

Communication and classroom conduct

Cyber Campus is the official location for announcements and course-wide questions. Students should participate respectfully, protect confidential or sensitive data, and avoid photographing or recording classmates without permission.

VI. Materials and References

Required materials will be provided through Cyber Campus and the course GitHub repository.

Primary tools

- Python in Google Colab; pandas, NumPy, matplotlib/seaborn, statsmodels, and scikit-learn.
- Git and GitHub for versioned, reproducible research workflows.
- Course repositories will be hosted under https://github.com/GSIS-DS.

Recommended open references

- McKinney, Wes. Python for Data Analysis, 3rd ed. Online edition: https://wesmckinney.com/book/
- Booth, Wayne C., et al. The Craft of Research, 4th ed.
- Creswell, John W., and J. David Creswell. Research Design: Qualitative, Quantitative, and Mixed Methods Approaches, 6th ed.
- Huntington-Klein, Nick. The Effect: An Introduction to Research Design and Causality: https://theeffectbook.net/
- VanderPlas, Jake. Python Data Science Handbook: https://jakevdp.github.io/PythonDataScienceHandbook/

Additional readings, datasets, methodological notes, and selected empirical studies will be assigned by topic.

Ⅶ. Course Schedule

(* Subject to change)

Week	Contents	Note
1	Tue: Course orientation: what research is; qualitative, quantitative, mixed-methods, descriptive, explanatory, causal, and predictive studies Thu: Research workflow and setup: Colab, GitHub, notebooks, file organization, and reproducibility	
2	Tue: From topic to researchable problem: questions, theory, concepts, units of analysis, hypotheses or propositions, feasibility, and contribution Thu: Question clinic: narrow an individual topic and build a preliminary research map	
3	Tue: Finding and evaluating literature: scholarly databases, search terms, citation chains, source quality, reference management, and responsible use of AI Thu: Literature-search lab: documented search, source verification, and annotated evidence matrix	

Week	Contents	Note
4	Tue: Writing a literature review: synthesis rather than summary; themes, debates, gaps, conceptual frameworks, and argument structure Thu: Synthesis workshop: literature map, gap statement, and proposal peer review	
5	Tue: Introduction to qualitative research: purposes, epistemological choices, case studies, interviews, focus groups, observation, and document analysis Thu: Qualitative design studio: sampling, protocols, access, consent, ethics, and reflexivity	
6	Tue: Qualitative evidence and analysis: transcription, coding, categories, themes, interpretation, trustworthiness, triangulation, and mixed-methods integration Thu: Coding lab with interview or document data; qualitative research exercise	
7	Tue: Quantitative research design: measurement, operationalization, surveys, sampling, secondary data, selection, missingness, and threats to validity Thu: Design clinic: align the individual question, literature, data, variables, and method; midterm review	
8	*Midterm Exam*	
9	Tue: Finding, evaluating, and documenting quantitative data: provenance, metadata, licensing, APIs, secondary sources, and validation Thu: Data-source lab: import, inspect, document, and validate the project dataset	
10	Tue: Reproducible data preparation and descriptive research: tidy data, joins, missing values, summary statistics, exploratory analysis, and visual evidence Thu: Cleaning and visualization studio: build a transparent raw-to-analysis pipeline	
11	Tue: Bivariate analysis and inference refresher: comparisons, association, sampling variation, confidence intervals, tests, effect sizes, and substantive significance Thu: Simulation and inference lab; select evidence appropriate to the research question	
12	Tue: Regression as a research method: specification, interpretation, uncertainty, diagnostics, and common threats to validity Thu: Regression lab: functional form, residuals, influential observations, and robust checks	
13	Tue: Beyond the baseline: causal reasoning, prediction versus explanation, and text as data Thu: Method-choice lab and AI verification: test code, claims, citations, and output sensitivity	
14	Tue: Writing the complete research paper: integrating the introduction, literature review, methods, results, discussion, conclusion, and limitations Thu: Full-draft workshop: structured peer review, reproducibility check, and presentation preparation	Full draft
15	Tue: Individual research presentations and Q&A — Group 1 Thu: Individual research presentations and Q&A — Group 2	
16	Tue: Individual research presentations and Q&A — Group 3 Thu: Final paper revision workshop and submission of the individual research project	Final project

VIII. Assessment Arrangements

The midterm practical, individual/group assignments, and final examination assignment are scheduled as shown above. Make-up assessments or deadline adjustments are available only in cases recognized by University rules. Students should contact the instructor as early as reasonably possible when an approved circumstance may affect participation or submission.

IX. Accessibility and Student Support

Sogang University is committed to creating a learning environment that meets the needs of its diverse student body. If you anticipate or experience any barriers to learning in this course due to your challenges, please feel welcome to discuss your concerns with me. Course standards remain the same, while reasonable adjustments will be implemented in accordance with University policy.

Ver. 1.0 (July 27, 2026)

*** The Fine Print: This syllabus and course schedule may be adjusted in response to student learning needs, data availability, or the instructor's judgment.**